4.4.2 Congruence of Line Segments

Measure and observe the following line segments.



Fig. 4.18

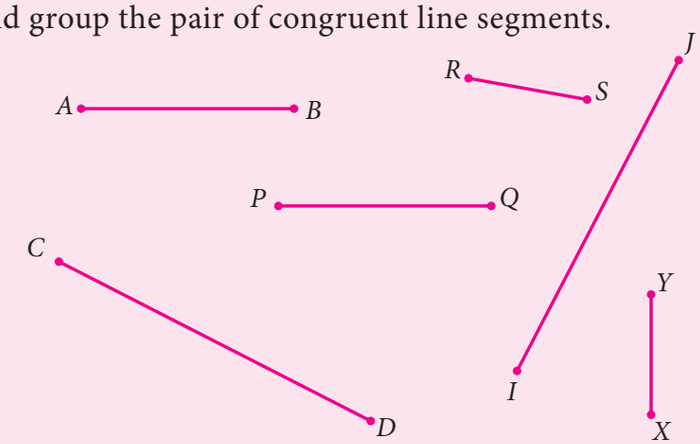
The line segments $\overline{AB}$ and $\overline{CD}$ have the same length. If we use the superposition method $\overline{AB}$ and $\overline{CD}$ will match each other. Hence the line segments are congruent and we can write it as $\overline{AB} \cong \overline{CD}$.

Since we are considering the lengths of the line segments for congruence we can also write $\overline{AB} \cong \overline{CD}$ as $\overline{AB} = \overline{CD}$.

From the above cases, we understand that lines are congruent if they have the same length.

Try these

Measure and group the pair of congruent line segments.



4.4.3 Congruence of Angles

Look at the following angles.

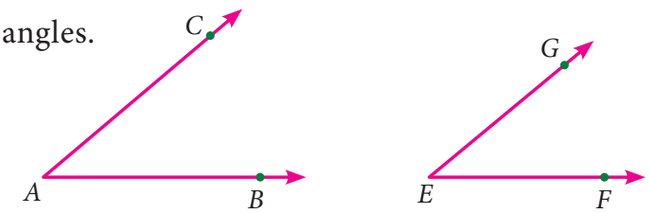


Fig. 4.19

To check the congruency of angles $\angle BAC$ and $\angle FEG$, make a trace copy of $\angle BAC$. By superposition method, if we try to match $\angle BAC$ with the $\angle FEG$ by placing the arm AB on the arm EF, then the arm AC will coincide with the arm EG. Even if length of the arms are different, the angles $\angle BAC$ matches exactly with $\angle FEG$. Hence the angles are congruent.

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